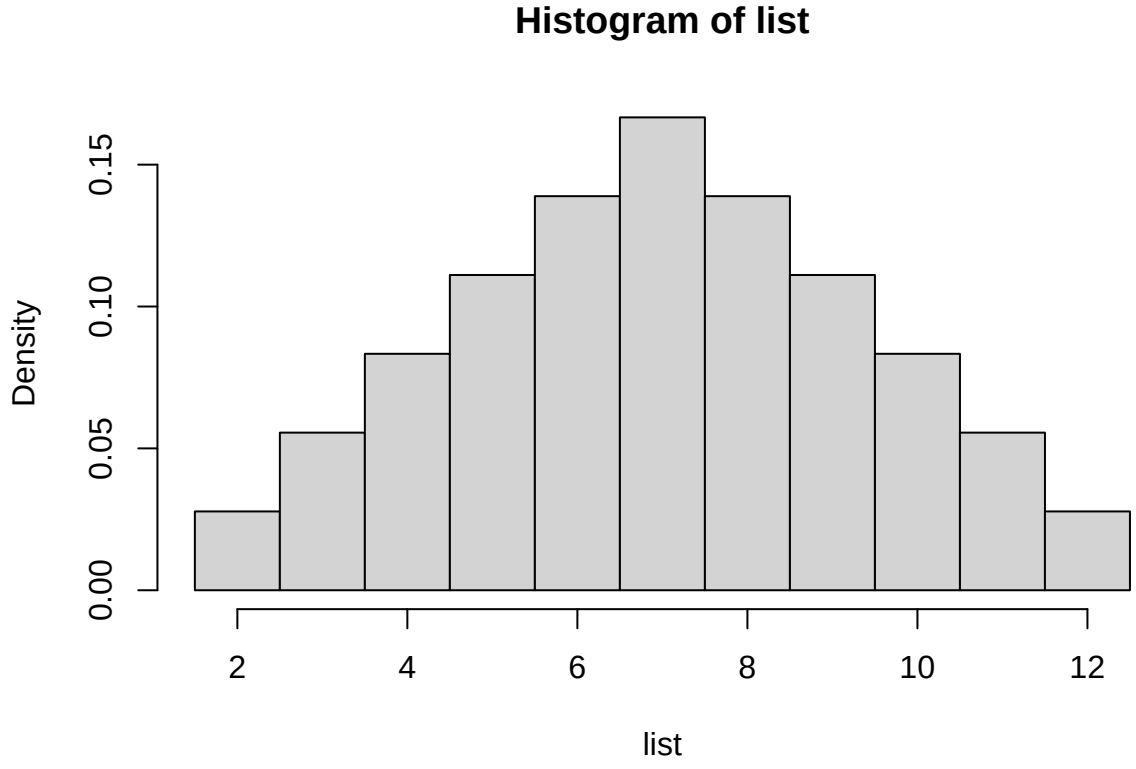
Solutions to Review Questions for Lecture 13

1.

Z	$P(Z)$
2	$1/36$
3	$2/36 = 1/18$
4	$3/36 = 1/12$
5	$4/36 = 1/9$
6	$5/36$
7	$6/36 = 1/6$
8	$5/36$
9	$4/36 = 1/9$
10	$3/36 = 1/12$
11	$2/36 = 1/18$
12	$1/36$

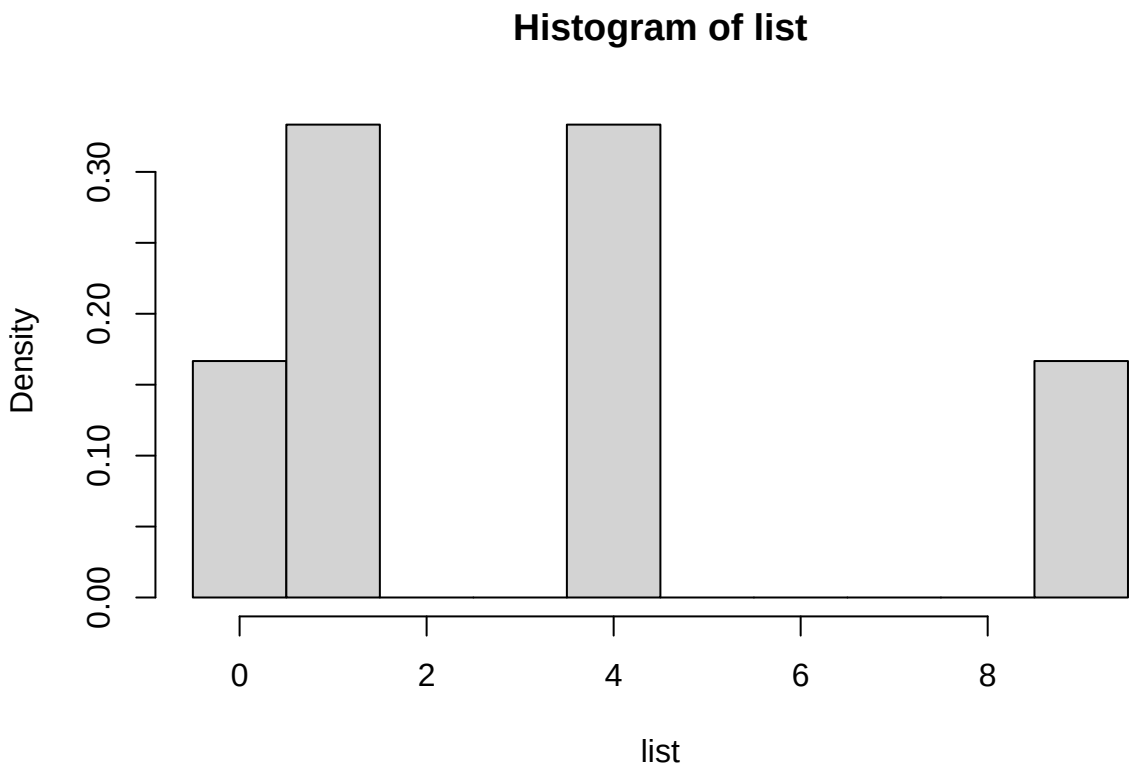
```
list=c(2,rep(3,2),rep(4,3),rep(5,4),rep(6,5),rep(7,6),rep(8,5),rep(9,4),rep(10,3),rep(11,2),12)
hist(list,freq=FALSE,breaks=seq(1.5,12.5,1))
```



2.

$Z = X^2$	$P(Z)$	X
0	1/6	3
1	2/6	2, 4
4	2/6	1, 5
9	1/6	6

```
list=c(0,1,1,4,4,9)
hist(list,freq=FALSE,breaks=seq(-0.5,9.5,1))
```



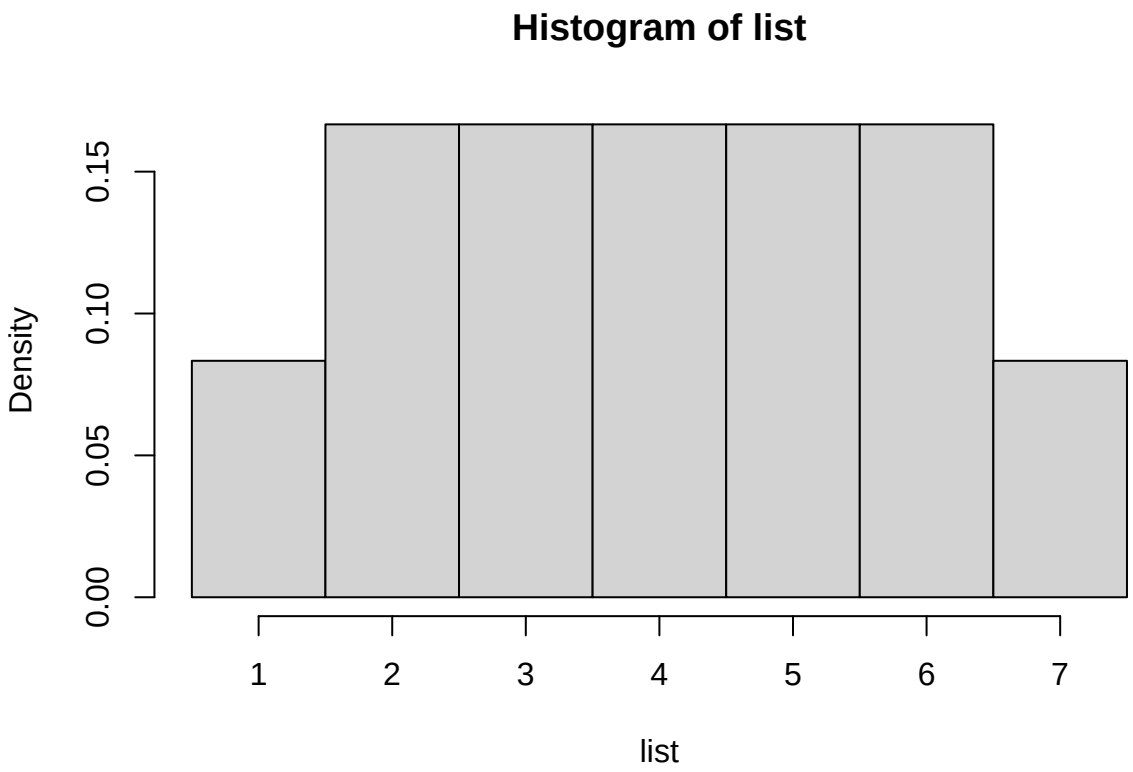
3.

X	$P(X)$	Y	$P(Y)$	$Z = X + Y$	$P(Z)$
1	1/6	1	1/2	2	$1/6 * 1/2 = 1/12$
1	1/6	0	1/2	1	$1/6 * 1/2 = 1/12$
2	1/6	1	1/2	3	$1/6 * 1/2 = 1/12$
2	1/6	0	1/2	2	$1/6 * 1/2 = 1/12$
3	1/6	1	1/2	4	$1/6 * 1/2 = 1/12$
3	1/6	0	1/2	3	$1/6 * 1/2 = 1/12$
4	1/6	1	1/2	5	$1/6 * 1/2 = 1/12$
4	1/6	0	1/2	4	$1/6 * 1/2 = 1/12$
5	1/6	1	1/2	6	$1/6 * 1/2 = 1/12$
5	1/6	0	1/2	5	$1/6 * 1/2 = 1/12$
6	1/6	1	1/2	7	$1/6 * 1/2 = 1/12$
6	1/6	0	1/2	6	$1/6 * 1/2 = 1/12$

Tabulating these results gives:

Z	$P(Z)$
1	1/12
2	$2/12 = 1/6$
3	$2/12 = 1/6$
4	$2/12 = 1/6$
5	$2/12 = 1/6$
6	$2/12 = 1/6$
7	1/12

```
list=c(1,2,2,3,3,4,4,5,5,6,6,7)
hist(list,freq=FALSE,breaks=seq(0.5,7.5,1))
```



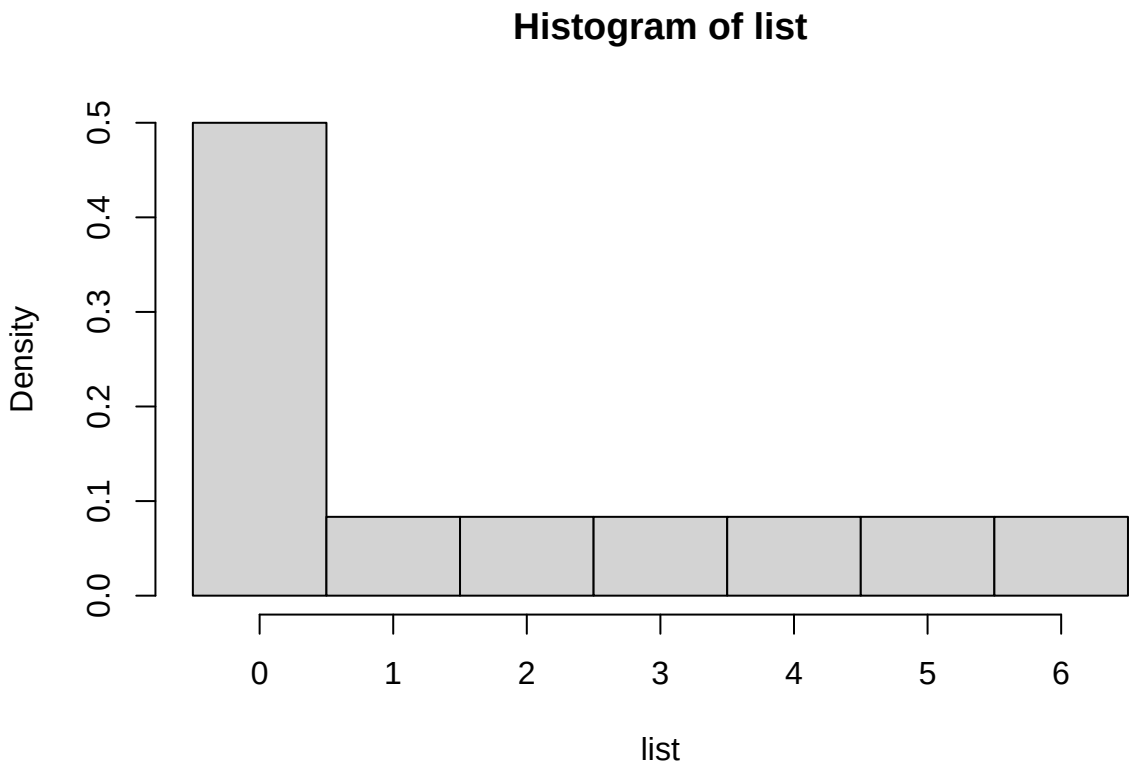
4.

X	$P(X)$	Y	$P(Y)$	$Z = X * Y$	$P(Z)$
1	1/6	1	1/2	1	$1/6 * 1/2 = 1/12$
1	1/6	0	1/2	0	$1/6 * 1/2 = 1/12$
2	1/6	1	1/2	2	$1/6 * 1/2 = 1/12$
2	1/6	0	1/2	0	$1/6 * 1/2 = 1/12$
3	1/6	1	1/2	3	$1/6 * 1/2 = 1/12$
3	1/6	0	1/2	0	$1/6 * 1/2 = 1/12$
4	1/6	1	1/2	4	$1/6 * 1/2 = 1/12$
4	1/6	0	1/2	0	$1/6 * 1/2 = 1/12$
5	1/6	1	1/2	5	$1/6 * 1/2 = 1/12$
5	1/6	0	1/2	0	$1/6 * 1/2 = 1/12$
6	1/6	1	1/2	6	$1/6 * 1/2 = 1/12$
6	1/6	0	1/2	0	$1/6 * 1/2 = 1/12$

Tabulating these results gives:

Z	$P(Z)$
0	$6/12 = 1/2$
1	$1/12$
2	$1/12$
3	$1/12$
4	$1/12$
5	$1/12$
6	$1/12$

```
list=c(rep(0,6),1,2,3,4,5,6)
hist(list,freq=FALSE,breaks=seq(-0.5,6.5,1))
```



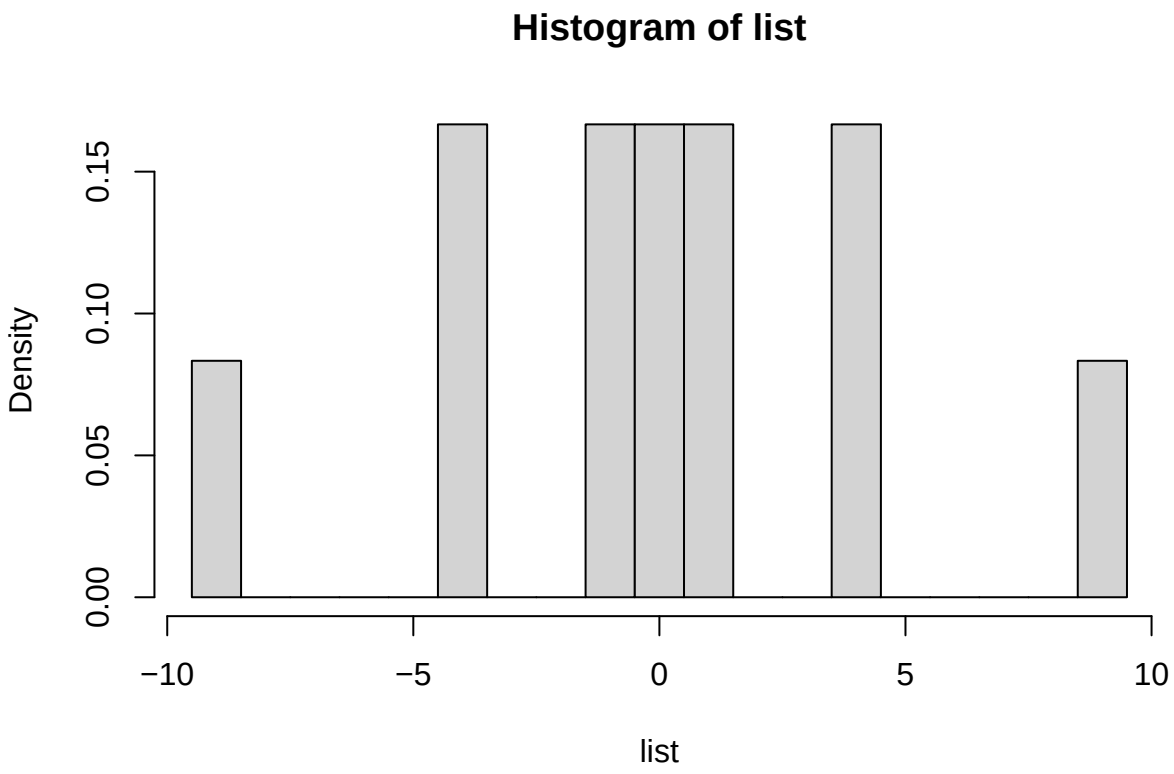
5.

X	$P(X)$	Y	$P(Y)$	$Z = (X-3)^2(2Y-1)$	$P(Z)$
1	1/6	1	1/2	4	$1/6 * 1/2 = 1/12$
1	1/6	0	1/2	-4	$1/6 * 1/2 = 1/12$
2	1/6	1	1/2	1	$1/6 * 1/2 = 1/12$
2	1/6	0	1/2	-1	$1/6 * 1/2 = 1/12$
3	1/6	1	1/2	0	$1/6 * 1/2 = 1/12$
3	1/6	0	1/2	0	$1/6 * 1/2 = 1/12$
4	1/6	1	1/2	1	$1/6 * 1/2 = 1/12$
4	1/6	0	1/2	-1	$1/6 * 1/2 = 1/12$
5	1/6	1	1/2	4	$1/6 * 1/2 = 1/12$
5	1/6	0	1/2	-4	$1/6 * 1/2 = 1/12$
6	1/6	1	1/2	9	$1/6 * 1/2 = 1/12$
6	1/6	0	1/2	-9	$1/6 * 1/2 = 1/12$

Tabulating these results gives:

Z	$P(Z)$
0	$2/12 = 1/6$
1	$2/12 = 1/6$
-1	$2/12 = 1/6$
4	$2/12 = 1/6$
-4	$2/12 = 1/6$
9	1/12
-9	1/12

```
list=c(rep(c(1,-1,4,-4,0),2),9,-9)
hist(list,freq=FALSE,breaks=seq(-9.5,9.5,1))
```



6.-7.

X	$P(X)$	Y	$P(Y)$	$X+Y$	$P(X+Y)$
1	1/3	2	1/2	3	$1/2 * 1/3 = 1/6$
1	1/3	4	1/2	5	$1/2 * 1/3 = 1/6$
3	1/3	2	1/2	5	$1/2 * 1/3 = 1/6$
3	1/3	4	1/2	7	$1/2 * 1/3 = 1/6$
5	1/3	2	1/2	7	$1/2 * 1/3 = 1/6$
5	1/3	4	1/2	9	$1/2 * 1/3 = 1/6$

Tabulating these results gives:

$X+Y$	$P(Z)$
3	1/6
5	$2/6 = 1/3$
7	$2/6 = 1/3$
9	1/6

8. $E(X) = 1 * 1/3 + 3 * 1/3 + 5 * 1/3 = 3$.
9. $E(X+Y) = 3 * 1/6 + 5 * 1/3 + 7 * 1/3 + 9 * 1/6 = 6$.
10. $SE(X) = \sqrt{(1-3)^2 * 1/3 + (3-3)^2 * 1/3 + (5-3)^2 * 1/3} = \sqrt{8/3}$.
11. $SE(X+Y) = \sqrt{(3-6)^2 * 1/6 + (5-6)^2 * 1/3 + (7-6)^2 * 1/3 + (9-6)^2 * 1/6} = \sqrt{11/3}$.

12.

Y	Y^2	$P(Y^2)$
2	4	1/2
4	16	1/2

13. $P(Y^2 = 4) = 1/2$.
14. $E(Y^2) = 4 * 1/2 + 16 * 1/2 = 10$.
15. $SE(Y^2) = \sqrt{(4-10)^2 * 1/2 + (16-10)^2 * 1/2} = 6$.
16. $E(Y) = 2 * 1/2 + 4 * 1/2 = 3, E(Y)^2 = 9$.
17. $\sqrt{E(Y^2) - E(Y)^2} = \sqrt{10-9} = 1$. Note that this is $SE(Y)$.
18. $SE(Y) = 1$.

19.

X	$P(X)$	Y	$P(Y)$	$Z = X * Y$	$P(Z)$
1	1/3	2	1/2	2	$1/2 * 1/3 = 1/6$
1	1/3	4	1/2	4	$1/2 * 1/3 = 1/6$
3	1/3	2	1/2	6	$1/2 * 1/3 = 1/6$
3	1/3	4	1/2	12	$1/2 * 1/3 = 1/6$
5	1/3	2	1/2	10	$1/2 * 1/3 = 1/6$
5	1/3	4	1/2	20	$1/2 * 1/3 = 1/6$

Tabulating these results gives:

$X+Y$	$P(Z)$
2	1/6
4	1/6
6	1/6
10	1/6
12	1/6
20	1/6

20. $P(Z = 12) = 1/6$.
21. $E(Z) = (2 + 4 + 6 + 10 + 12 + 20)/6 = 9$.
22. $SE(Z) = \sqrt{E(Z^2) - E(Z)^2}$. We know that $E(Z^2) = (2^2 + 4^2 + 6^2 + 10^2 + 12^2 + 20^2)/6 = 350/3$, and so $SE(Z) = \sqrt{350/3 - 9^2} = \sqrt{107/3}$.

23.

X	$P(X)$	Y	$P(Y)$	$U = (Z-3)(Y-3)$	$P(U)$
1	1/3	2	1/2	2	$1/2 * 1/3 = 1/6$
1	1/3	4	1/2	-2	$1/2 * 1/3 = 1/6$
3	1/3	2	1/2	0	$1/2 * 1/3 = 1/6$
3	1/3	4	1/2	0	$1/2 * 1/3 = 1/6$
5	1/3	2	1/2	-2	$1/2 * 1/3 = 1/6$
5	1/3	4	1/2	2	$1/2 * 1/3 = 1/6$

Tabulating these results gives:

U	$P(U)$
0	$2/6 = 1/3$
-2	$2/6 = 1/3$
2	$2/6 = 1/3$

24. $E(U) = 0 * 1/3 + -2 * 1/3 + 2 * 1/3 = 0$. This is expected since $E(U)$ is the covariance between X and Y, which are independent.

25. $Cov(X, Y) = E(U) = 0$.

26. We know that $Cov(X, X^2) = E(X * X^2) - E(X) * E(X^2) = E(X^3) - E(X) * E(X^2)$. We know further that $E(X) = (-4+0+4)/3 = 0$, $E(X^2) = (16+0+16)/3 = 32/3$, and $E(X^3) = (-64+0+64)/3 = 0$. Hence $Cov(X, X^2) = 0-0*32/3 = 0$. This is significant because it is an example of two non-independent random variables with 0 covariance.

27. a) We know that $V = \frac{\sum_{i=1}^{n}(X_i-\bar{X})^2}{n} = \frac{\sum_{i=1}^{n} X_i^2-2X_i\bar{X}+\bar{X}^2}{n} = \frac{\sum_{i=1}^{n} X_i^2-2\bar{X}\sum_{i=1}^{n} X_i+n\bar{X}^2}{n}$. We also know that $\sum_{i=1}^{n} X_i = n\bar{X}$, and so the previous expression simplifies to $\frac{\sum_{i=1}^{n} X_i^2-2\bar{X}n\bar{X}+n\bar{X}^2}{n} = \frac{\sum_{i=1}^{n} X_i^2}{n} - \bar{X}^2$. Hence $E(V) = E(\frac{\sum_{i=1}^{n} X_i^2}{n} - \bar{X}^2) = \frac{E(\sum_{i=1}^{n} X_i^2)}{n} - E(\bar{X}^2) = \frac{\sum_{i}^{n} E(X_i^2)}{n} - E(\bar{X}^2)$. The random variables X_i are identically distributed, and so $E(X_i^2) = E(X_1^2)$ for all i. The last expression then simplifies to $E(V) = E(X_1^2) - E(\bar{X}^2)$.

    b) We know that for all i, $Var(X_i) = SE^2(X_i) = \sigma^2$ and also that $Var(X_i) = E(X_i^2) - E(X_i)^2 = E(X_i^2) - \mu^2$. Combining these two equations and solving for $E(X_i^2)$ gives $E(X_i^2) = \sigma^2 + \mu^2$ for all i.

    c) We know that $\bar{X}^2 = \frac{(\sum_{i=1}^{n} X_i)^2}{n^2}$. When we square this sum, there will be n X_i^2 terms and $n^2 - n$ terms that are X_iX_j with $i \neq j$. The X_i^2 terms will have expected value $\sigma^2 + \mu^2$ from part b) and the X_iX_j terms will have expected value $E(X_iX_j) = E(X_i)E(X_j) = \mu^2$ because X_i and X_j are independent if $i \neq j$. It follows that $E(\bar{X}^2) = \frac{n(\sigma^2+\mu^2)+(n^2-n)(\mu^2)}{n^2} = \frac{n\sigma^2+n^2\mu^2}{n^2} = \mu^2 + \frac{\sigma^2}{n}$.

    d) Combining the results from a), b), and c) gives $E(V) = E(X_1^2) - E(\bar{X}^2) = \sigma^2 + \mu^2 - (\mu^2 + \frac{\sigma^2}{n}) = \sigma^2(\frac{n-1}{n})$.